

Wireless Sensor Networks in Agriculture Through Machine Learning: A Survey

Why it is relevant:

This paper is relevant because it explores the integration of machine learning with wireless sensor networks for agricultural monitoring and automation. Since your project involves smart irrigation and intelligent decision-making, the paper provides valuable insights into AI-based farming solutions.

Summary:

The paper provides a comprehensive survey of how machine learning techniques are applied to wireless sensor networks in agriculture. It examines the use of sensor data collected from soil moisture, temperature, humidity, and other environmental parameters to support intelligent farming decisions. The study discusses various machine learning algorithms used for crop monitoring, irrigation management, disease detection, yield prediction, and anomaly detection. By combining WSNs with machine learning, agricultural systems can analyze field conditions more accurately, automate decision-making processes, and improve resource utilization. The paper concludes that AI-driven sensor networks have significant potential to enhance productivity, efficiency, and sustainability in modern agriculture.

Relation to Your Project:

- Wireless sensor networks (WSN) ✓
- Machine learning integration ✓
- Soil moisture monitoring ✓
- Smart irrigation support ✓
- Real-time data analysis ✓
- Intelligent decision-making ✓
- Smart farming automation ✓
- Precision agriculture applications ✓